UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	INTERACTIVE MULTIMEDIA
COURSE CODE	:	CSC413
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC413 mock paper (NOV 2026).
Answers show the expected marking points; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 15 x 1 mark; Part B = 15 x 1 mark; Part C = 30 x 1 mark; Part D = 4 x 10 marks; total = 100.

4. Questions: 64. Slots: 72. Placeholders: 0. No LLM call.
Questions and answers are for local study use only.

5. Structure derived from 15 unique CSC413 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A - ANSWER KEY

Q	Ans	Statement	Explanation
A1	FALSE	A linear presentation allows users to choose their own path.	A linear presentation follows a fixed sequence.
A2	TRUE	Multimedia combines text, graphics, audio, video and animation.	These are the core multimedia elements.
A3	FALSE	A plotter is an input device.	A plotter produces output.
A4	TRUE	A web browser displays hypertext documents.	Browsers render web pages.
A5	TRUE	Chroma key compositing replaces a colour background with another image.	Chroma key removes a selected colour.
A6	TRUE	Hypermedia combines multimedia with hyperlinks.	Hypermedia adds linked navigation to multimedia.
A7	TRUE	HTML is used to structure content on web pages.	HTML marks up web content.
A8	TRUE	A scanner is an input device.	A scanner captures documents into the computer.
A9	FALSE	MIDI files store digitized audio samples.	MIDI stores musical instructions, not samples.
A10	FALSE	A multimedia project always uses every media element.	A project selects the elements it needs.
A11	FALSE	RAM is a type of permanent storage.	RAM is temporary working memory.
A12	FALSE	A hyperlink can link only to another web page, not to a file.	Links can target files and other resources.
A13	TRUE	A licence states how a work may be used.	A licence defines permitted use.
A14	FALSE	Interactivity level is irrelevant when choosing an authoring tool.	The required interactivity affects tool choice.
A15	TRUE	Card-based authoring tools are best for messages with a clear beginning and end.	Cards support sequential navigation.

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PART B - ANSWER KEY

Q	Ans	Term	Definition
B1	C	Keyframe	important frame in an animation
B2	G	Bitmap	image made of pixels
B3	F	Sampling	converting sound into digital values
B4	K	Serif	font with small strokes at character ends
B5	O	MIDI	standard for musical notes and instructions
B6	L	Sans serif	font without small strokes
B7	N	Vector	image made of mathematical objects
B8	A	Browser	software that displays web pages
B9	H	Storyboard	visual sequence plan
B10	D	Kerning	adjustment of space between character pairs
B11	J	Frame rate	frames displayed per second
B12	I	Amplitude	height of a sound wave
B13	E	Card-based tool	authoring tool organised like pages or cards
B14	B	Target audience	intended users of a product
B15	M	Tweening	generating in-between frames

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PART C - ANSWER KEY

Q	Ans	Correct answer
C1	C	A video compression family
C2	A	Card-based
C3	B	Modelling and rendering
C4	D	Frequency
C5	B	One shape into another
C6	C	Temporary working memory
C7	C	A simulated environment
C8	B	PNG
C9	C	Plotter
C10	C	Screen display
C11	A	An early model of a product
C12	B	Pixels
C13	A	Drawing each frame by hand
C14	B	Assemble multimedia elements
C15	D	In-between frames
C16	C	Text, graphics, audio, video and animation
C17	C	Play while downloading
C18	D	Along a predefined route
C19	A	Frames per second
C20	D	Merging of media and computing
C21	A	Scanner
C22	C	The user can control content and flow
C23	B	It combines visuals and sound
C24	C	CMYK
C25	D	Smooth jagged edges
C26	A	Multimedia with hyperlinks
C27	A	Draws alternate lines of a frame
C28	C	Improves accuracy
C29	B	A cinema presentation
C30	D	Vector

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PART D - WORKED ANSWERS

QUESTION 1

Topic: web hypermedia

A museum wants a website that presents its collection and lets visitors explore related exhibits.

a) State TWO differences between linear and non-linear navigation.

Linear follows a fixed sequence; non-linear lets users choose their own path.

(3 marks)

b) Describe TWO roles of hyperlinks in the website.

They connect related exhibits and let users move between pages.

(3 marks)

c) Explain how streaming and compression affect multimedia delivery on the web.

Compression reduces file size for faster transfer; streaming lets audio or video play while it downloads.

(4 marks)

QUESTION 2

Topic: text

Text is an essential element in a multimedia project. Its shape, size, design, colour, arrangement and linking all affect the message.

a) State TWO reasons sans serif is mostly used for screen display and serif is used for printed material.

Sans serif is cleaner at screen sizes; serif aids reading in print.

(2 marks)

b) Describe TWO roles of hypertext in multimedia.

It links related content and supports non-linear navigation.

(3 marks)

c) Explain how text can have a massive impact on an informative multimedia project in the link and HTML context.

Clear headings and links guide users through HTML pages and make the information easier to scan and follow.

(5 marks)

QUESTION 3

Topic: authoring interactivity

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

Interactivity support and export formats are important features.

(2 marks)

b) Explain the interactivity level suitable for the demonstration.

A medium level with menus and buttons lets users explore without overwhelming them.

(4 marks)

c) Explain TWO authoring tool categories appropriate for this project.

Card-based tools suit structured screens; timeline-based tools suit animation and media sequencing.

(4 marks)

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QUESTION 4

Topic: animation

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

GIF and MP4 are suitable; SWF is also acceptable.

(3 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

Draw each frame, capture the frames in sequence, then play them back; it gives direct artistic control.

(3 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

2-D uses flat drawings such as a cartoon; 2.5-D adds depth to flat artwork, such as parallax layers in a game.

(4 marks)

END OF MARKING SCHEME

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